

Post Agent

Human-in-the-loop long-form blog post generator (LangGraph pipeline)

Overview

The Post Agent generates deeply researched, 3000–4500 word long-form magazine-style blog posts. It uses a LangGraph StateGraph with 9 nodes that execute sequentially. After generation, the post is saved as 'pending' status and surfaced in the admin panel for human review. An admin must explicitly approve or reject each post before it goes live. Both CLI and admin UI approval are supported.

Invocation & Human-in-the-Loop Flow

Phase 1 — Generate	python -m meridian_agents.post_agent → runs full pipeline, saves as PENDING
Phase 2 — Approve	python -m meridian_agents.post_agent approve or admin UI
Phase 2 — Reject	python -m meridian_agents.post_agent reject or admin UI
Pending registry	post_agent/pending_posts.jsonl (one JSON entry per pending post)
SDK	LangGraph (langgraph + langchain-core)
Primary LLM	gemini-2.5-flash (via Gemini OpenAI-compatible API)
Fallback LLM	gpt-4o (OPENAI_API_KEY)
Image — Travel	Unsplash real photos (UNSPLASH_ACCESS_KEY)
Image — Other	Imagen 4 via Gemini → FLUX.1-dev (HF) → FLUX.1-schnell (HF) → Unsplash
Taxonomy	MCP server at /api/mcp — list_categories, list_tags, create_blog, manage_guest_user

LangGraph Node Pipeline

Node	What it does
init_mcp	Sends MCP initialize handshake to /api/mcp
ensure_author	Creates or verifies the AI guest author account via MCP manage_guest_user

Node	What it does
discover_trend	<code>_pick_category()</code> selects a category (AI/Technology/Science/History/Travel/Knowledge). <code>_web_search()</code> queries Gemini Search grounding (fallback: OpenAI <code>web_search_preview</code>) for the hottest topic.
deep_research	Fires 3 parallel web searches: technical details, community reactions, real-world implications. Uses <code>ThreadPoolExecutor(max_workers=3)</code> .
write_post	<code>chat_completion()</code> sends category persona + research to LLM. Returns JSON with title, excerpt, HTML content (with <code>[[IMAGE:...]]</code> placeholders), <code>featuredImagePrompt</code> , category/tag keywords. Minimum 3000 words targeted.
expand_post	Conditional node: if <code>word_count < 2200</code> , sends current JSON back to LLM with expansion prompt. Otherwise skipped.
generate_images	Generates featured image (1792×1024) and resolves all <code>[[IMAGE:...]]</code> placeholders to tags. Travel posts use Unsplash; others use Imagen 4 → FLUX.1-dev → schnell → Unsplash.
pick_taxonomy	Calls MCP <code>list_categories</code> and <code>list_tags</code> . Matches post keywords to existing taxonomy. Resolves <code>category_id</code> and up to 6 <code>tag_ids</code> .
save_pending	Calls MCP <code>create_blog</code> with <code>status='pending'</code> . Stores <code>pending_post_id</code> in state.
publish_approved	Only runs on the approve resume flow. Calls <code>PATCH /api/posts/:id/approve</code> .

Topic Categories & Research Modes

Category	Description
AI	<code>ai_trending</code> mode or random; uses Gemini Search for latest ML/AI news
Technology	general pool; hardware, software, cybersecurity, space, biotech
Science	general pool; physics, astronomy, biology, climate, medicine
History	general pool; lesser-known events and turning points
Travel	general pool; trending destinations; uses Unsplash for real photos

Category	Description
Knowledge	general pool; psychology, economics, philosophy, linguistics

TOPIC_MODE env var controls selection: empty=random, ai_trending=AI category, random_general=non-AI pool, or an exact category name.

Image Generation Pipeline

For Travel posts: Unsplash API (real photos). For all others:

Priority	Provider
1 — Gemini Imagen 4	gemini-2.5-flash enhances the prompt, then imagen-4.0-generate-001 renders it at target aspect ratio
2 — Gemini Imagen 4 Fast	imagen-4.0-fast-generate-001 as fallback for Imagen 4
3 — FLUX.1-dev (HF)	DSLR-quality prompt prefix + 30-step inference via HuggingFace Inference API
4 — FLUX.1-schnell (HF)	Fast HuggingFace model fallback
5 — Unsplash stock	Last resort: keyword search against Unsplash API

State Schema (AgentState TypedDict)

Fields: server_base, author_name/email/password, category_name/pool/research_style, trend, technical, reactions, implications, post_title, post_excerpt, post_content, post_featured_image_prompt, post_category_keywords, post_tag_keywords, post_unsplash_query, word_count, featured_image_url, final_content, category_id, tag_ids, pending_post_id, pending_post_slug, approved, published_slug, published_id.

Key Files

File	Purpose
post_agent/main.py	Phase 1 (generate) and Phase 2 (approve/reject) CLI. Pending registry management.
post_agent/graph.py	LangGraph StateGraph builder — 9 nodes, conditional edge, interrupt_before publish
post_agent/state.py	AgentState TypedDict definition
post_agent/nodes/research.py	discover_trend_node + deep_research_node; all web search logic
post_agent/nodes/writer.py	write_post_node + expand_post_node; per-category personas and prompts

File	Purpose
post_agent/nodes/images.py	generate_images_node; Imagen, FLUX, Unsplash providers; upload helper
post_agent/nodes/mcp.py	init_mcp, ensure_author, pick_taxonomy, save_pending, publish_approved nodes
post_agent/tracer.py	AgentRun lifecycle tracking
meridian_agents/llm.py	Shared chat_completion() with Gemini primary / gpt-4o fallback